

Financial News Sentiment Alpha Study

Rebuilt from DuckDB after dbt tests and a top-to-bottom notebook execution

FULL-SAMPLE DAILY MEAN

0.236%

132 days | HAC p=0.042 | block p=0.047

COMPOUNDED EW RETURN

34.1%

Before costs; 200% gross exposure and 0% net exposure.

FF3 + MOMENTUM ALPHA

0.207%

HAC p=0.099; 95% CI includes zero.

HOLDOUT DAILY MEAN

-0.027%

53 days | HAC p=0.849 | cumulative -1.6%

Decision-relevant conclusion

The positive full-sample association does not survive the chronological holdout and is not statistically significant after FF3+momentum adjustment. The OOS-minus-IS difference is -0.438% per day (block p=0.037). The evidence is therefore not reliable proof of tradable alpha.

Interpretation is before transaction costs, financing, short borrow, slippage and corporate-action adjustments.

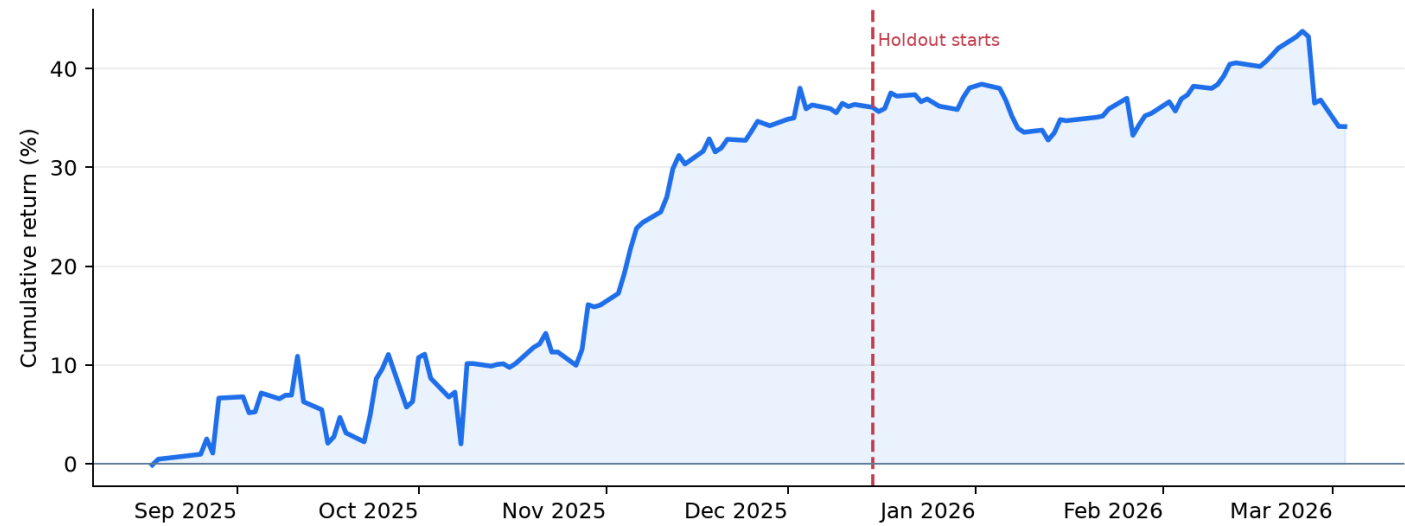
Sample and provenance

- Signal dates: 2025-08-18 to 2026-03-03
- Panel: 107,422 ticker-days across 6,632 tickers
- Realized return dates: 2025-08-19 to 2026-03-04
- News: 540,216 distinct articles / 781,542 ticker-article rows

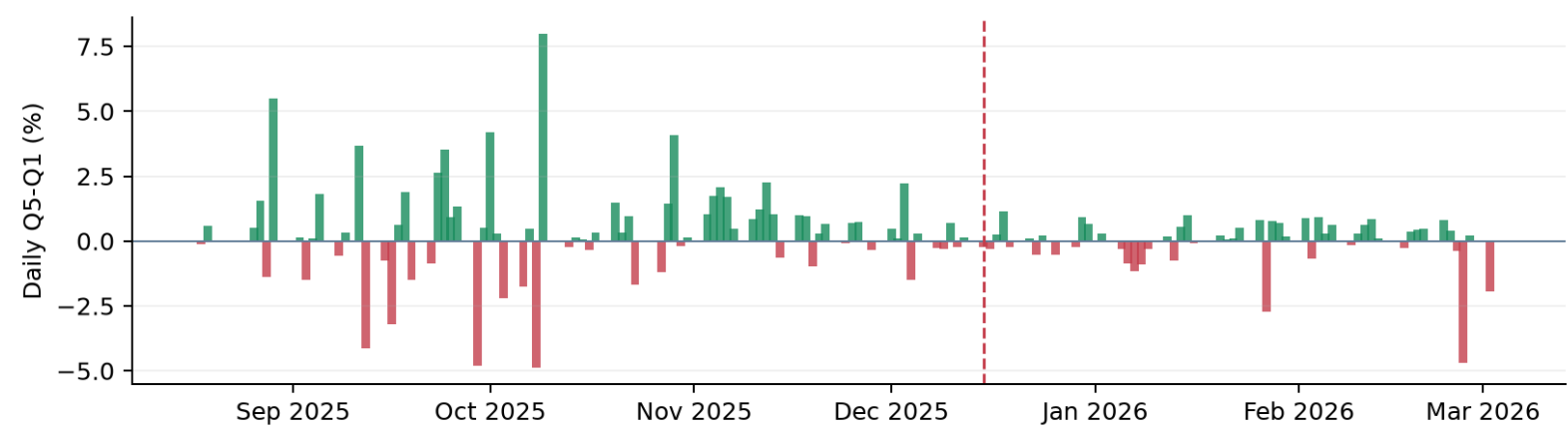
Performance path and stability

Canonical portfolio: signal at the 16:00 ET cutoff, close(date_t) to next common trading-session close

Compounded equal-weight long-short return



Daily Q5-minus-Q1 returns



Frozen chronological split

In sample

n=79 | mean 0.412% | HAC p=0.007

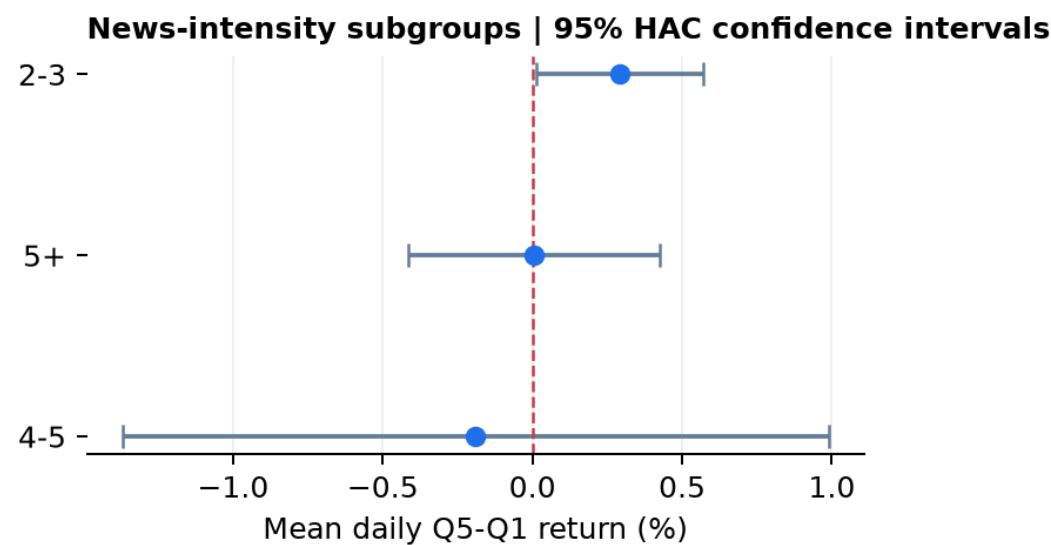
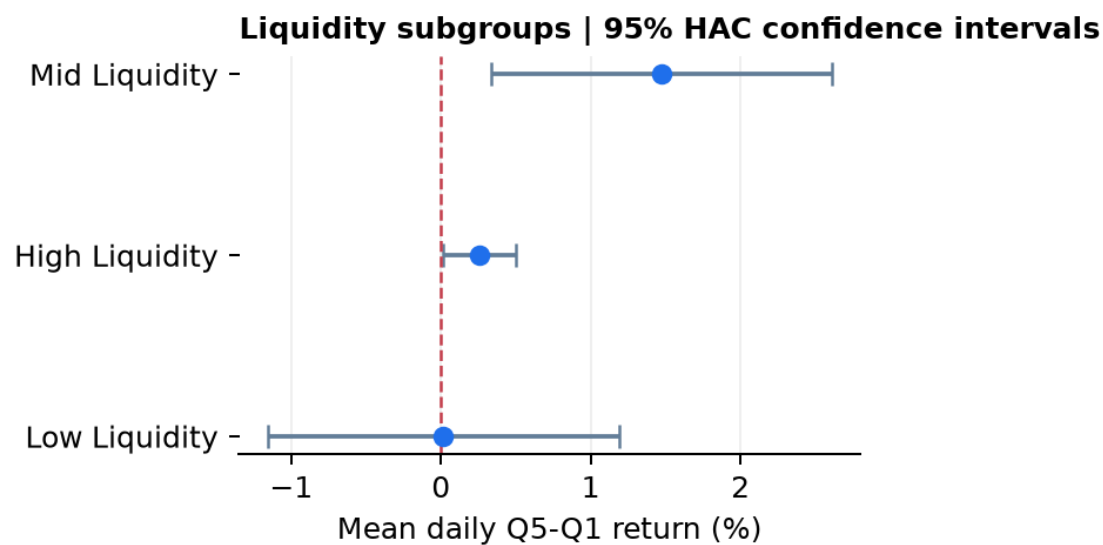
Holdout

n=53 | mean -0.027% | HAC p=0.849

Holdout - training: -0.438%/day

Subgroup analysis with dependence-aware inference

Each subgroup uses Newey-West HAC (5 lags), 5-day circular moving-block bootstrap and Holm correction



What the subgroup tests do — and do not — establish

- Mid-liquidity mean: 1.471%/day; Holm-adjusted HAC p=0.034.
- Low-minus-high liquidity contrast: -0.207%/day; HAC p=0.725.
- 2-3 article subgroup: 0.291%/day; Holm-adjusted HAC p=0.120.
- 5+-minus-2-3 article contrast: -0.380%/day; HAC p=0.059.

Conclusion: direct contrasts do not support a monotonic liquidity or news-intensity effect.

Definitions, data integrity and reproducibility

The report, notebook and Power BI template all read the same tested DuckDB model

Canonical definitions

Signal time

Articles before 16:00 ET map to that session; later items map to the next session.

Return horizon

close(date_t) to close(next common market session); factors join on the realized return date.

Daily portfolio

Equal-weight Q5 minus Q1; one unit long and one unit short.

Cumulative return

PRODUCT(1 + daily return) - 1; arithmetic spread is diagnostic only.

Inference and validation

HAC

Newey-West standard errors with five lags for means and factor alpha.

Bootstrap

5,000 circular moving-block draws, five-day blocks, fixed seed 42.

Factors

Daily FF3 plus momentum aligned to return_date_t.

Holdout

Frozen start 2025-12-15; no construction or inference rules change across the split.

Data health

Point-in-time universe

107,422/107,422 panel rows pass the price-history-based universe rule.

Current metadata

Only 71,668 rows have current company metadata; missing metadata does not exclude securities.

Coverage

Average daily signal coverage is 5.4%.

Pipeline checks

9 Python unit tests and 72 successful dbt nodes (11 models + 61 data tests).